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SUPERSEDING
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FEDERAL SPECIFICATION

CLAMP, ELECTRICAL, (LIVE LINE CONNECTOR)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a connector, otherwise known as a "hot line clamp", which joins a branch conductor to the main conductor in such a way that the connector can be attached or detached by means of an insulated stick.

1.2 Classification.

1.2.1 *Types and classes.* Live line connectors furnished under this specification shall be of the following types and classes, as specified (see 6.1).

Type I—For general applications.
Type II—For application in industrial atmosphere.
Class 1—For copper-to-copper connections.
Class 2—For aluminum-to-aluminum connections.
Class 3—For aluminum-to-copper connections.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:

PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-636—Box, Fiberboard.
PPP-B-665—Boxes, Paperboard, Metal Stayed, (Including Stay Material).
PPP-B-676—Boxes, Set-up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-116—Preservation, Methods Of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

FSC 5975

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MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

American Society for Testing And Materials (ASTM) Standard:

B117—Salt Spray (Fog) Testing, Standard Method of.

(Copies may be obtained from the American Society for Testing And Materials, 1916 Race Street, Philadelphia, Penna.)

National Electrical Manufacturer's Association (NEMA) Standard:

SG1-1962—Electric Power Connectors, Standards for.
SG14-1958—Electrical Connectors for Copper Conductors, Standards for.

(Copies may be obtained from the National Electrical Manufacturers Association, 155 East 44th Street, New York, N. Y.)

Edison Electric Institute (EEI) Standard:

TDJ-162—Overhead Distribution Connectors for Aluminum Conductors, Standards for.

(Copies may be obtained from the Edison Electric Institute, 750 Third Avenue, New York, N. Y.)

3. REQUIREMENTS

3.1 General. Live line connectors shall be constructed of a metal or metals which, when the device is installed under conditions of actual service and exposed to moisture, will not be liable to be affected adversely by electrolysis. Their design shall be such that they will have adequate strength and rigidity to permit installation in the intended manner without rupture or distortion which would adversely affect service life, and will carry the current which they may be called upon to handle without overheating. The means for attachment of the tap conductor shall be such that this conductor is securely and reliably held.

3.2 Material.

3.2.1 *Class 1 clamps.* Class 1 clamps shall be fabricated from a copper alloy capable of meeting the performance requirements of 3.3.

3.2.2 *Class 2 clamps.* Class 2 clamps shall have heat-treated aluminum bodies capable of meeting the performance requirements of 3.3.

3.2.3 *Class 3 clamps.* Class 3 clamps shall have heat-treated aluminum bodies and high strength bronze terminals and shall be capable of meeting the performance requirements of 3.3.

3.3 Performance.

3.3.1 *Temperature rise.* The temperature rise of the connectors shall conform to the requirements of NEMA SG1-1962, paragraph 3.04.

3.3.2 *Resistance.* The connector resistance shall conform to the requirements of NEMA SG1-1962, 5.02C.

3.3.3 *Thermal cycling.* The ability of the connectors to handle the thermal expansion and contraction of the conductors shall conform to the requirements of EEI TDJ-162, section I, paragraphs 4.1 through 4.2.

3.3.4 *Corrosion resistance.* The connector shall meet the requirements of 3.3.1 and 3.3.2 after exposure to a salt fog atmosphere as set forth in 4.4.4.

3.3.4.1 *Industrial atmospheres (type II connectors only).* The connector shall meet the requirements of 3.3.1 and 3.3.2 after exposure to a hydrogen sulfide atmosphere as set forth in 4.4.5.

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3.3.5 *Mechanical properties.*

3.3.5.1 *Class 1 clamps.* Class 1 clamps shall conform to the requirements of NEMA SG14-1958, part 3.

3.3.5.2 *Classes 2 and 3 clamps.* Classes 2 and 3 clamps shall conform to the requirements of EEI TDJ-162, section I, paragraphs 4.3 through 4.5.

3.4 Workmanship. Workmanship shall be first class throughout. Items shall be free from burrs, sharp or rough edges, or any defects which may impair their serviceability or durability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Sampling for visual examination shall be in accordance with MIL-STD-105, inspection level II, acceptable quality level (AQL) 6.5.

4.2.2 Sampling for test shall be in accordance with MIL-STD-105, inspection level S2, AQL 6.5.

4.3 Inspection. Samples selected in accordance with 4.2.1 shall be subjected to a visual examination to determine compliance with the requirements of 3.1 and 3.4.

4.4 Tests. Samples selected in accordance with 4.2.2, shall be subjected to the following tests to determine compliance with the applicable requirements of 3.3.

4.4.1 *Temperature rise.* Live line connectors shall be tested in accordance with paragraph 5.01 of NEMA SG1-1962 to determine their temperature rise characteristics (see 3.3.1).

4.4.2 *Resistance.* Connector resistance shall be determined in accordance with paragraph 5.02 of NEMA SG1-1962 (see 3.3.2).

4.4.3 *Thermal cycling.* The test to determine thermal expansion and contraction characteristics of the connectors shall be conducted in accordance with section II, paragraph B of EEI TDJ-162 (see 3.3.3).

4.4.4 *Corrosion test.* Connectors shall be tested as follows to determine their ability to withstand corrosion (see 3.3.4):

(a) Attach the connector as recommended by the manufacturer to 24 inches of the largest size stranded cables for which the connector is designed, using a torque of 175 inch-lbs. for both the eyescrew and jumper terminals.

(b) Subject the connector assembly to the salt spray test set forth in ASTM B117 for a period of 1000 hours.

(c) Test the connector to determine compliance with 3.3.5.

4.4.5 *Hydrogen sulfide test.* Type II connectors shall be tested as follows to determine their resistance to attack by industrial atmospheric contaminants (see 3.3.4.1):

(a) Attach the connector as recommended by the manufacturer to 5 inches of the largest size stranded cables for which the connector is designed, using a torque of 175 inch-lbs. for both the eyescrew and jumper terminals.

(b) Mix 196 grams of Na_2SO_3 with 97 grams of Na_2S in a desiccator, add 293 cc of distilled water, and stir thoroughly.

(c) Place connector assembly in desiccator, add 392 cc of hydrochloric acid (HCl-Sp. Gr. 1.19) to above solution, and close desiccator.

(d) After exposure of the assembly to the hydrogen sulfide for a period of 500

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hours, test the connector to determine compliance with 3.3.5.1.

4.4.6 *Mechanical tests.*

4.4.6.1 *Class 1 clamps.* Class 1 clamps shall be subjected to the tests set forth in NEMA SG14-1958, paragraphs 4.04 through 4.06 to determine compliance with 3.3.5.1.

4.4.6.2 *Classes 2 and 3 clamps.* Classes 2 and 3 clamps shall be subjected to the tests set forth in section II, paragraph C, of EEI TDJ-162 to determine compliance with 3.3.5.2.

4.5 Examination of preparation for delivery requirements. An examination shall be made to determine that the packaging, packing, and marking comply with the requirements of section 5 of this specification. Defects shall be scored in accordance with table I. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be II and the AQL shall be 4.0 defects per hundred units.

TABLE I. *Classification of preparation for delivery defects*

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate sealing. Bulging or distortion of container.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and application of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.1).

5.1.1 *Level A.* Each clamp shall be packaged in accordance with MIL-P-116, method IC-2 or IC-5.

5.1.2 *Level B.* Each clamp shall be packaged in a box conforming to PPP-B-566, style II, type A, class a; PPP-B-665, style optional; or to PPP-B-676, type and class optional.

5.1.3 *Level C.* The clamps shall be packaged in accordance with the suppliers commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 *Level A.* The clamps, packaged as specified in 5.1, shall be packed in a box conforming to PPP-B-636, V3c. The box shall be closed, waterproof sealed with tape, and reinforced with bands of filament tape in accordance with the appendix to the box specification.

5.2.2 *Level B.* Unless otherwise specified (see 6.1) 48 clamps shall be packed in a box conforming to PPP-B-636, type CF, class domestic. Closure shall be in accordance with the appendix to the box specification.

5.2.3 *Level C.* The clamps shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to markings required by the contract or order, the interior packages and shipping contain-

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ers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered hereon and procurement documents should specify the following:

(*a*) Title, symbol, and date of this specification.
(*b*) Type and class of clamp (1.2.1).
(*c*) Conductor sizes (main and jumper) to be accommodated.
(*d*) Inspection responsibility, if other than specified (4.1).
(*e*) Packaging and packing levels required (5.1 and 5.2).
(*f*) Special marking, if required (5.3).

6.2 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Insulator Clamps, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor:

Cable Clamps, not otherwise indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY INTERESTS:

NONE

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